

Intrusion Detection System on IoT Devices via Federated Learning

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Abstract—In the rapidly changing Internet of Things (IoT) landscape, IoT devices that control our everyday lives are becoming a primary target for intruders. It has tended to become more vulnerable and exposed to handling a diverse range of data and functions. Traditional centralized intrusion detection systems failed to address its robustness against the scale and diversity of IoT networks. Furthermore, centralized ML models pose significant privacy concerns because sensitive data transfers to a central server, which opens up potential data leakage. To address these challenges, we propose Federated Learning-based Intrusion Detection System (FLIDS). Our FLIDS not only preserves privacy but also minimizes the resources required for data transmission. With FLIDS, real-time learning and adaptation to new attacks are always possible because IoT devices can continuously update their models based on local changes in network data without waiting for batch processing, which is typically required in traditional ML models. Moreover, FLIDS has been implemented using Federated Proximal to address the heterogeneous data environment, and different training configurations have been performed to tune the model parameters. Our model achieved a higher accuracy of 99.31% and an f1-score of 97.38%, underscoring the excellent detection performance.

Index Terms—Internet of Things (IoT); Intrusion Detection System (IDS); Federated Learning (FL); Machine Learning (ML); Neural Networks; Privacy; Security

I. INTRODUCTION

Due to rise of Internet of Things (IoT) has heavily transformed the way we live and work, creating an expansive network of connected devices that offer innovative applications across various sectors. IoT devices that equipped with sensors and internet connectivity, have introduced undeniable convenience and efficiency in domains such as healthcare, transportation, home automation, and industrial operations. In healthcare, IoT enables remote patient monitoring and advanced diagnostics, improving patient outcomes and reducing healthcare costs. In transportation, smart traffic management systems and connected vehicles enhance safety and reduce congestion. Home automation systems provide comfort and energy efficiency, while industrial IoT streamlines operations

and boosts productivity. However, this widespread connectivity also brings critical security challenges. IoT devices raises significant amount of concerns about data privacy as it generates and transmits vast amount of data. Unauthorized access to sensitive information can lead to severe privacy breaches. Addressing these security issues is essential to harness the full potential of IoT while safeguarding data and maintaining the reliability of interconnected systems. Improving adversary detecting tools for identifying and evaluating malicious behaviors in IoT networks is essential due to emerging threats. It is a essential ability to analyze big data for processing the vast and diverse information generated by IoT devices in real-time. [1] There are three main categories of malicious or intrusion detection [2]: anomaly detection, signature detection, and hybrid detection. Basically in the anomaly detection approach, the system learns about the benign network behavior and identifies any types of deviations from it and identify that as potential attacks. In signature detection approach, on the other hand, system compares the captured events with signatures or patterns which was already stored, to detect known attack types patterns. Both anomaly and signature detection methods are combined in hybrid detection approach and it aims to take strengths of each for more robust detection capabilities.

In the traditional ML-based method, the entire dataset kept as centralized data, which may be resulted in lengthy training times to identify deviations if any [3]. This centralized behavior also raise privacy issues regarding data. Also communication cost can increase, there could be scalability problems as number of IoT devices is growing over the period. However, sometimes multi-view approach considers multiple sets of features, where efficiency can be improved through selecting proper features. As a result, attack detection will be more accurate.

Federated Learning (FL) is an approach where training is conducted by utilizing decentralized data, all while maintaining the privacy appropriately [4]. At present, FL has gained

popularity as a solution to preserve privacy in edge devices by providing low latency updates and enabling knowledge sharing among devices under same FL network.

Our study focuses on using Federated Learning techniques to detect intrusion on IoT devices. This paper proposes a Federated Learning-based Intrusion Detection System (FLIDS) aimed at detect intrusions in decentralized approach. The need to transfer raw data to a central server is eliminated which helps in preserving privacy as data remained local. Data leakage is minimized by this approach while maintaining model accuracy. We evaluated FLIDS, which is aided by CNN, using a recent network dataset and it yields superior performance compared to other existing methods. We incorporated Federated Proximal, a aggregation algorithm in Federated Learning, into the FLIDS to enhance the performance across diverse network settings. Then we tested trained model on a different network dataset to assess its adaptability and robustness.

The remaining sections of this paper are organized as follows: Chapter II provides an overview of related research. Chapter III outlines the entire methodology in a step-by-step manner, including the hypertuned model and the Federated Learning technique used to detect intrusions. Chapter IV presents the results and evaluates the performance of the Federated Learning approach. Finally, the conclusion is presented in Chapter V.

II. RELATED WORK

In this section, we analyze the existing works based on intrusion detection systems in both traditional and federated learning methods.

The Canadian Institute for Cybersecurity (CIC) [5] published a study that conducted the network traffic behavior of 60 IoT devices under various conditions (on, inactive, in-use, interacting) using WiFi, ZigBee, and Z-Wave protocols. They simulated real-world smart home scenarios with benign data flows and seven common cyberattacks. The researchers trained machine learning classifiers (Decision Tree, Random Forest, XGBoost, AdaBoost) on the generated dataset, achieving the highest accuracy with a Decision Tree model.

Similarly, this paper [6] presented a newly profiled dataset IoTID20. It was also a reference point to identify anomalous activity. They divided the dataset in binary and multiclass. The dataset got 80 features and 3 labels named binary, category, and subcategories. They evaluated different types of machine learning classifiers where regression, LDA, Gaussian NB, and SVM performed poorly. Decision tree classifier showed the best accuracy of 88%. The Bot-IoT dataset [7] encompasses various types of cyber attack types which incorporate both simulated and real-world IoT network traffic. It reflects realistic scenarios like smart thermostats and motion-activated garage doors. The creators implemented feature selection techniques to identify the most informative ones for intrusion detection. They trained model on a subset of the most relevant features using recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) and achieved 99% accuracy. In case of entire feature set, Support Vector Machine (SVM) comes out as

the best model. Yang and wang [8] proposed an improved convolutional neural network (ICNN) for intrusion detection in wireless networks. Their method demonstrated enhanced detection accuracy and true positive rate maintaining a lower false positive rate that tested on intrusion behavior. The study [9] exhibits the evolution of intrusion detection systems by contrasting the classification performance of kNN and DNN and stresses the significance of handling current datasets when assessing intrusion detection strategies. DNN outperformed KNN in terms of accuracy, precision, and recall like DNN has accuracy of 96.427%, in compared to kNN's 90.913%, also DNN had better response time. The paper didn't discuss the real world scenario or any comprehensive analysis. Ferrag et al. [10] introduced a deep learning-based intrusion detection system (IDS) to detect DDoS attacks in Agriculture 4.0. The IDS utilizes three deep learning models : convolutional neural networks (CNN), deep neural networks (DNN), and recurrent neural networks (RNN). To assess the performance, the study evaluated the TON IoT dataset and the CICDDoS2019 dataset. Upon examining the results of both datasets, they obtained 99% for the binary class on all three DL models and 95%, 94%, and 93% for the multiclass (13 classes) on CNN, RNN, and DNN, respectively. Also, the TON IoT dataset achieved nearly 99% accuracy in Multiclass.

FedAGRU, [11] an improved federated learning algorithm that combined with Gated Recurrent Unit (GRU) and Support Vector Machine (SVM) to suggest an intrusion detection model. The aggregation technique of FedAGRU assigns varying weights to different clients. Compared to FedAVG, fewer communication rounds and more than 90% accuracy have been achieved using FedAGRU. FedAGRU reduced the communication cost by 70% in comparison to other Federated Learning methods. Researchers proposed an AI-driven intrusion detection system [12] using federated learning for IoT networks. They designed a high-level architecture that leverages virtual IoT instances and a specific deep learning model (GRU) with variable window and layer sizes. Their evaluation compared the federated learning approach to a non-federated one. Their architecture achieved 91% accuracy outperforming non-federated approach that results in 86% accuracy. However, limited range of attack types are considered in this study, whereas real-world scenarios involve a much broader spectrum of potential attacks. However, a limitation of this study is that it only considered a limited range of attack types, whereas real-world scenarios involve a much broader spectrum of potential attacks. Attota et al. [13] introduced MV-FLID, a FL technique that detects different classes of attacks on edge IoT devices by training on multiple views of network data. Based on packet information three distinct views were developed, named Bi-Directional Features (Bi-flow view), Uni-Directional Features (Uniflow view), and Packet-Features (Packet view). They detected adversary nodes with an accuracy of 94.172%. Suggested method can be useless when significant processing power is needed for security gateways. In this related study [14], Industrial Internet of Things (IIoT) environments were used to detect intrusion using Federated

Learning (FL). Evaluating an Edge-IIoTset dataset, for centralized models, they used CNN and RNN, and for three different types of client sets, they used FL models. Additionally, they evaluated the model on two sets of data called IID and non-IID. They worked on three different client sets $k=3, 9, 15$ and evaluated the result on two different FL rounds 1st and 50th, for CNN model best and worst client Accuracy were % 90.19%, 58.23% while global model Accuracy was and 90.73% and in case of RNN they got 91.06%, 74.71% and 91.07% respectively, after 50th FL rounds and when $k=15$. Li et al. [15] proposed a FL-based NIDS named DAFL that assigned weights on local client models based on performance. Also, they attempted to lower the communication overhead. The model was evaluated with different client set sizes, and the results showed the following performance metrics: when three clients were involved the accuracy, precision, recall, and F1-score were 94.64%, 93.61%, 92.87%, 93.24%; when Client numbers were 5, they were 94.61%, 93.44%, 92.98%, 93.21%; and in Client, number was equal to seven, they were 94.58%, 93.38%, 93.22%, 93.3%.

Main limitations are network data heterogeneity. Data heterogeneity introduction can cause client models to deviate from global model which can cause slower convergence. Our model can address this as FedProx will introduce a regularization term that controls how much each local model can deviate from the global model and stabilize the training process in heterogeneous network environment.

III. METHODOLOGY

In this section, we will first explain our CNN model and then analyze the Federated Learning framework.

A. Convolutional Neural Network

We use Convolutional Neural Network (CNN) that has characteristics of extracting features. We employ CNN as both local and global deep learning model. The architecture shown in figure1

1) *Convolutional Layer*: We applied conv1D where convolutional works like a filter. We employed a filter sizes of 32 to 128 and kernel sizes of 3 & 5.

2) *Pooling Layer*: The pooling layer summarizes each grid square into a single value which reduces the complexity. We have used maxpooling which selects the maximum value of the grid. Our maxpooling size was two.

3) *Fully Connected Layer & Dropout*: In Fully Connected Layer, neurons are interconnected to the previous layer's neurons and it allows the network to identify higher-level relationships. Our experimented dense units ranged between 32 to 128. Dropout is a regularization way to drop a certain percentage of neurons in the layer randomly and prevent overfitting. The rate in our model is varied between 0.3 to 0.5.

4) *Activation Functions*: Activation function introduce non-linearity in neural network. In 1D convolution and fully connected layer we used ReLu and for output layer we used sigmoid function.

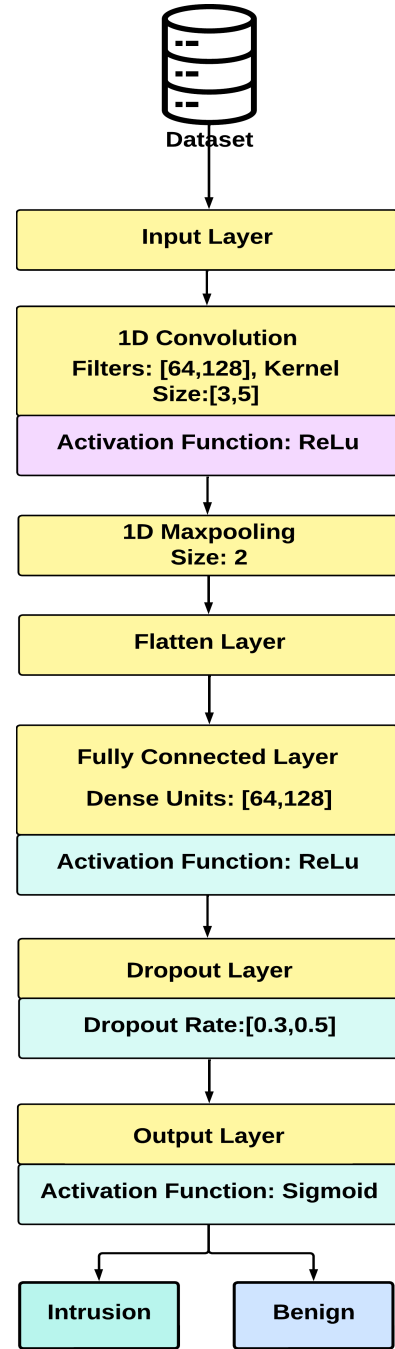


Fig. 1. Our CNN model architecture

- **ReLu**: It stands for rectified linear unit that set negative or zero value to zero otherwise the value is unchanged.

$$R(z) = \max(0, z) \quad (1)$$

- **Sigmoid**: It is basically a mathematical function which can map any value between 0 to 1.

$$f(x) = \frac{1}{1 + e^{-x}} \quad (2)$$

5) *Training Setup*:: We performed hyper-parameter tuning to optimize our global model parameters. A table of all parameter list is given below:

Parameters	Values
Trials	30
Direction	Minimize (Validation Loss)
Epochs	10
Batch Size	128
Optimizer	Stochastic Gradient Descent (SGD)
Loss Function	Binary Crossentropy
Learning Rate	$[1e-5 \text{ to } 1e-2]$
Momentum	$[0.0 \text{ to } 0.9]$

TABLE I
HYPERPARAMETER TUNING FOR SERVER MODEL SELECTION

B. Federated Learning based IDS:

Federated Learning (FL) is emerging as a potential approach to train ML models on decentralized data. This approach allows connecting local clients devices collaboratively to learn from a global model without directly sharing their sensitive raw data. Our main framework on federated learning depicted in figure 2. The steps that are involved in implementing FL are given below:

1) *Generation of Client Models*: The main foundation in FL is to create client model. We created a Five clients set to see how consistently our model perform in different local client model. Then we cloned the global model into those five clients. At last, we evaluated each client model after integration and aggregation with global server to see how robustly it is performing.

2) *Initialization*: First, the global server initializes the global model and local clients downloads the selected parameters. The central model serves as the initial reference for local training on each client's dataset.

3) *Local Training*: Using provided dataset, each local clients independently trains the received global model on their local data which results in a locally updated model. This training is typically performed for a predefined parameters like epochs, batch size, or other model parameter.

4) *Aggregation Scheme*: To get a refined global model, central server aggregates local updates provided by clients. We use FedProx as aggregation scheme.

- **Federated Averaging (FedAvg)**: During Aggregation in central server, federated averaging strategy is used. Here local weights from local clients are aggregated based on dataset size [16]. The Federated Averaging (FedAvg) update equation is given by:

$$v_{t+1} = \sum_{i=1}^K \frac{n_i}{N} z_t^{(i)} \quad (3)$$

where:

- t : Communication round.

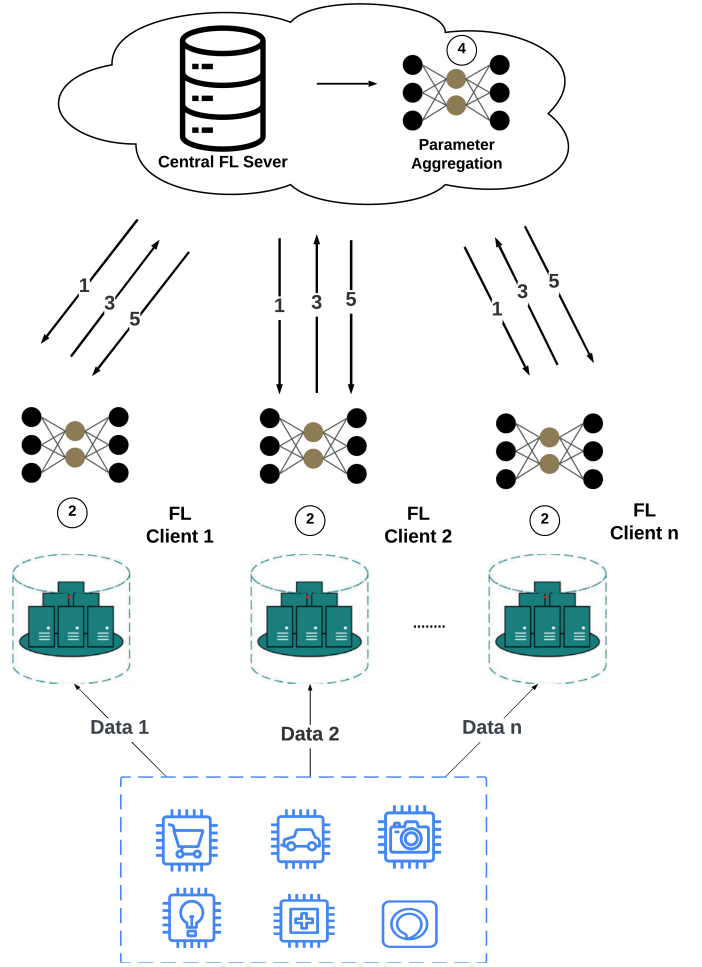


Fig. 2. Methodology of FL-based IDS. 1. Generation of initial model ; 2. Local model training and update ; 3. Local model updated parameters upload; 4. Server aggregation and Global model parameters update; 5. Distribute the updated global model parameters; Repeat 2-5, until model convergence;

- v_{t+1} : Updated global model parameters after aggregation in round $t + 1$.
- K : Number of participating clients.
- N : Total data points of all participating clients, given by $N = \sum_{i=1}^K n_i$.
- n_i : Number of data points for client i .
- $z_t^{(i)}$: Model parameters from client i after local optimization with the proximal term.

- **Federated Proximal (FedProx)**: FedProx [17], an extension of FedAvg, adds a regularization term during local training which lessen the effect of data heterogeneity. The proximal term addition to standard loss function helps to maintain consistency over performance of clients. We used FedProx instead of FedAvg so that FLIDS performs better in different network data. The modified loss function in FedProx is represented as:

$$L_i(w) = G_i(z_t^{(i)}, y_i) + \frac{\lambda}{2} \|z_t^{(i)} - v_t\|^2 \quad (4)$$

where:

- $L_i(w)$: Modified loss function for client i .
- $G_i(z_t^{(i)}, y_i)$: Local loss calculation for client i .
- $z_t^{(i)}$: Model parameters of client i before optimization.
- v_t : Global model parameters received by the client.
- λ : Regularization parameter controlling the strength of the proximal term.
- $\frac{\lambda}{2} \|z_t^{(i)} - v_t\|^2$: Proximal Term
- y_i : Input data of client i .

5) Upload, Aggregate and Distribute Update:

- Once training is complete, the central server have the updated weights of client models.
 - The server then apply weighted average approach similar to Federated average during aggregation of received local updates. However, in this case, data heterogeneity handled more effectively due to the proximal term.
 - Finally, the updated global model is sent back to local clients, who incorporate the new weights for the next round of training.
- The whole process is circularly repeated until the model converges.

6) *Training Setup For FL Communication Round:* Training configuration that we used for Federated Learning communication rounds is given in table-II.

Parameters	Values
Communication Rounds	10
Participating Clients	5
λ (FedProx)	0.01
Learning Rate	0.0096
Momentum	0.483
Epochs	10
Batch Size	64
Loss Function	Binary Crossentropy
Optimizer	Stochastic Gradient Descent (SGD)

TABLE II

FOR ALL COMMUNICATION ROUNDS TRAINING SETTING

IV. EXPERIMENTAL EVALUATION

A. Dataset:

In evaluating the FL approach, we have used a real time dataset was produced by integrating large scale attack named CIC IoMT dataset 2024 [18]. To accomplish the dataset 18 types of attacks were executed on 40 IoMT devices and these attacks were further classified into seven main categories named DDoS, DoS, Recon, MQTT, and spoofing.

B. Centralized IDS:

In this section, we implemented centralized training to simulate a scenario. Here data from multiple clients is combined into a single dataset for model training. While this centralized approach yields strong detection performance, it introduces significant privacy concerns. And FLIDS server model not

only performs better (table-III) but also provides necessary client privacy and addressing risks associated with centralized system.

Model	Accuracy	Precision	Recall	F1-Score
Traditional IDS	99.27	96.20	98.39	97.26
FLIDS	99.31	96.22	98.62	97.38

TABLE III

PERFORMANCE OF CENTRALIZED DETECTION SYSTEM

C. FL Performance:

Earlier we evaluated our model using five clients. All five clients achieving robust results and our server model also keeping the consistency client models as we can observe in table-IV It underscore the benefit of FL as our server model outperforming client model and makes it a compelling solution in intrusion detection.

Model	Accuracy	Precision	Recall	F1-Score
Client 1	99.23	95.78	98.39	97.05
Client 2	99.15	96.45	96.97	96.71
Client 3	99.28	96.14	98.55	97.31
Client 4	99.32	96.40	98.51	97.43
Client 5	99.29	96.31	98.31	97.28
Server Model	99.31	96.22	98.62	97.38

TABLE IV

PERFORMANCE EVALUATION OF 5 CLIENTS AND SERVER MODEL

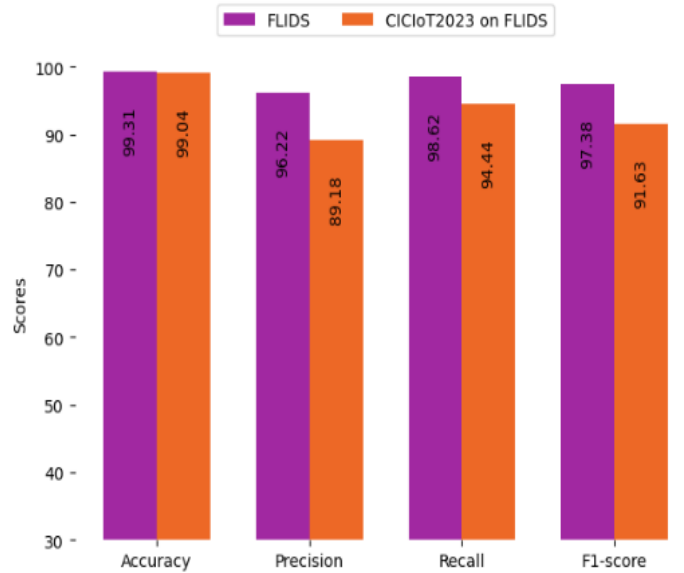


Fig. 3. Performance comparison between CICIOT 2024 and CICIOT 2023

D. Model Evaluation using Different Dataset:

To observe our already trained model performance in different network environment, we have used a real time dataset was produced by integrating large scale attack named CICIOT2023 [11]. To accomplish the dataset 33 attacks were executed on 105 IoT devices. To test the model's adaptability and robustness in a more dynamic environment, we selected the dataset. FLIDS demonstrated strong performance in terms of accuracy, meeting expected standards for intrusion detection. In other three additional performance metrics, it achieved a comparable result, indicating consistent and reliable detection capabilities (Fig-3). This evaluation focused on the server-side model that serves as a baseline for future assessments in more distributed settings.

E. Comparison with existing works:

We compared our proposed model with DAFL Dynamic Weighted Aggregation Federated Learning [15] and MV FLID Multi-View Federated Learning for Intrusion Detection [13]. The results in fig-4 indicate FLIDS not only provides a higher detection accuracy but also achieves a better balance between precision, recall and f1 score, are essential metrics for intrusion detection.

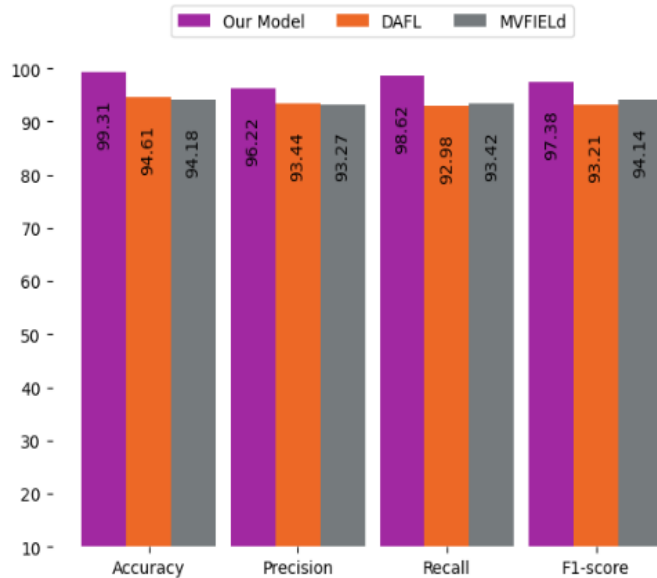


Fig. 4. Performance comparison with existing works

V. CONCLUSION

Federated Learning (FL) represents a significant advancement in machine learning by enabling decentralized training while preserving data privacy. In this work, we presented an Intrusion Detect System using Federated Learning and our model used federated proximal (fedprox) so that our FLIDS perform better in different network settings. We tuned our model by experimenting in different training configuration to identify optimal settings. We have given a detailed design of FLIDS and model performance was effective. It not only

surpassing centralized ML system but also performing better in compare to many existing methods achieving 99.31% accuracy and F1-score of 97.38%. In the future, we will focus on implementing model averaging with local fine-tuning and work on communication overhead. Additionally, we will explore security vulnerabilities in Federated Learning, such as model poisoning and adversarial attacks, and investigate mitigation strategies.

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